

Amrutesh Arun

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WORK EXPERIENCE

RENCI, University of North Carolina – Chapel Hill

Web Application Developer Intern

May 2025 – Present

Chapel Hill, NC (USA)

- Developed a **Node.js/Express/TypeScript** microservice backend with **PostgreSQL (Prisma)** for ROBOKOP web application, implementing GitHub, Google, and passkey-based authentication to enable users to save and load queries.
- Integrated authentication into a **React** frontend with protected routes, profile management, and query UI, while improving developer workflow through hot-reload support and bug fixes.
- Upgraded the entire codebase from React v16 to v19 and changed the codebase from JavaScript to TypeScript.
- Containerized the backend using **Docker** for consistent development and production environments.

University of North Carolina at Charlotte

Graduate Research Assistant | October 2024 - Present

August 2024 – August 2025

Charlotte, NC (USA)

- Assisted in researching and developing a **Large Language Model** based HIV awareness **mobile application** in collaboration with UNC School of Nursing, Yale University and Duke University.

Graduate Teaching Assistant | August 2024 - Present

Charlotte, NC (USA)

- Assisted in delivering course content, focusing on full-stack development using **NodeJS, EJS** and **MongoDB**.
- Developed an entire **React** course from scratch, which included creating videos on React, assignments, and projects.

Wasserstoff

Software Developer

August 2022 – November 2023

Gurugram, Haryana (IN)

- Responsible for development of 10+ cross-platform application in **blockchain** and **e-commerce** sectors, reaching a user base of 10,000+, collaborated with UI/UX designers to create **SEO-optimized pages**, boosting organic traffic by 20%.
- Created responsive web and mobile apps using **React, NextJS, React Native** and **Expo** frameworks Engineered scalable **APIs** and real-time features using **NodeJS, ExpressJS**, and **Socket.IO**, facilitating seamless trading and currencies.

upGrad Jeet

Software Developer

August 2021 – July 2022

Bangalore, Karnataka (IN)

- Developed & maintained admin panel, revamped the test engine utilized by over 5,000 educators and learners.
- Integrated error tracking and performance monitoring with **Sentry**, reducing user-reported issues by 40%.
- Implemented user engagement tracking via MoEngage & Google Analytics, driving improvements based on data insights.

EDUCATION

University of North Carolina- Charlotte

Master's of Science in Information Technology

January 2024 – Present

Charlotte, NC (USA)

Visvesvaraya Technological University

Bachelor of Engineering in Information Science

August 2017 – July 2021

Bangalore, Karnataka (IN)

PROJECTS

WorkQuest ([Project Link](#))

- A comprehensive redesign of an existing job search platform inspired by Craigslist Jobs, aimed at improving user experience, accessibility, and functionality (**Figma, JavaScript, React, Tailwind, Node, MongoDB**).

NYC Airport Traffic Analysis ([Project Link](#))

- Analyzed 20+ years of passenger & freight to identify seasonal trends (**Python, Pandas, NumPy, Matplotlib, Plotly**)

PayShare ([Project Link](#))

- A platform designed to simplify expense sharing among friends and groups. It helps users to split bills, track expenses, and settle debts effortlessly. (**TypeScript, React Native, Redux Toolkit, Node, Express, MongoDB**)

CERTIFICATIONS, SKILLS & INTERESTS

- Skills:** JavaScript, TypeScript, HTML, CSS, Java, Python, C, NoSQL, SQL, React, Next.js, Node.js, Flask, Django, React Native, Expo, Redux, Tanstack, Tailwind, Material UI, Framer, Jest, AWS, GraphQL, Figma, Postman, GitHub, VS Code
- Certifications:** [AWS Certified Solutions Architect – Associate](#), [Google IT Automation with Python](#), [Foundation of UX](#)
- Achievements:** Won 3 hackathons related to AI, LLMs, Web Development, UI/UX and Automation.
- Paper Publications:** “Design of Hierarchical Data Aggregation for Internet of Underwater Things” at ICGCCP-2021.